

The Hypothesis of Continuous Gravitational Contraction as a Foundation for the Decreasing Universe Theory: Redshift, Cosmological Time, Rotation Curves, and Observational Tests

Abstract

This article presents a synthetic formulation of the **Decreasing Universe Theory** — DUT — based on the propositions, equations, data, and observational tests contained in the documents analyzed. The theory's central hypothesis is that the gravitational field causes a continuous contraction of space and of all objects within it, including measuring instruments. This contraction would be extremely slow on human scales but cumulative on cosmological scales, potentially producing observational effects normally interpreted as accelerated expansion of the universe, dark energy, dark matter, cosmological redshift, and time dilation. From this hypothesis, the documents derive relationships between apparent distance, comoving distance, and redshift; propose an explanation for the time dilation observed in Type Ia supernovae; formulate an alternative for galactic rotation curves; derive the Tully-Fisher relation; and propose observational tests involving galactic mass, morphological age, and redshift. The results presented in the appendices indicate that, in some datasets, more massive or older galaxies would tend to exhibit lower redshift under comparable conditions. However, the documents themselves also indicate relevant limitations: small sample size, reliance on stellar mass as an approximation of total mass, projection effects, peculiar velocities, and the need for greater mathematical formalization. It is concluded that DUT, as analyzed here, constitutes an alternative theoretical program that is internally coherent in its unifying ambition, but still dependent on independent empirical validation and rigorous quantitative comparison with competing cosmological models.

Keywords: Decreasing Universe Theory; redshift; gravitational contraction; dark energy; dark matter; Tully-Fisher; Type Ia supernovae; alternative cosmology.

1. Introduction

The Decreasing Universe Theory — DUT — starts from a simple physical proposition: gravitational fields would not merely curve or affect space, but would cause a **continuous contraction of space and of everything within it**. In the documents analyzed, this contraction includes material objects, atomic standards, and measuring instruments. Thus, an observer situated within a gravitational field would not locally perceive their own contraction, since their rulers, clocks, and physical standards would contract along with them. The difference would become observable only in cosmological comparisons between regions subjected to distinct gravitational histories.

In the appendices, DUT is presented as an interpretive alternative to phenomena that, in the standard cosmological model, are usually explained by cosmic expansion, dark energy, dark matter, peculiar velocities, or relativistic effects. The theory proposes that the apparent accelerated recession of galaxies could result from the progressive contraction of local distance standards. If terrestrial instruments gradually become smaller, extragalactic distances would be measured as progressively larger, producing the impression of accelerated recession.

The same hypothesis is extended to redshift. The longer a photon's travel time to Earth, the longer the period during which the observer and their instruments would have continued to contract; therefore, the received wavelength would be measured as relatively longer. Redshift, therefore, would not depend only on distance, but also on the mass of the emitting galaxy, the galaxy's age, and the peculiar dynamics of the observed system.

This article organizes the propositions from the attached documents into scientific format, with emphasis on four elements: **hypothesis**, **methodology**, **theoretical model / calculation**, and **conclusion**. The goal is not to assert the definitive validation of DUT, but to reconstruct its internal logic and indicate how its data and equations are mobilized to support an alternative cosmological hypothesis.

2. Hypothesis

The general hypothesis can be formulated in the following terms:

Main hypothesis:

The gravitational field causes a continuous contraction of space, matter, and local measurement standards. This contraction is cumulative over time and proportional to the intensity of the gravitational field. On cosmological scales, this effect could produce redshift, time dilation, an apparent increase in distances, anomalous galactic rotation curves, and empirical relations usually attributed to the expansion of the universe, dark energy, or dark matter.

This hypothesis unfolds into four specific propositions.

The first proposition is that **the apparent distance of extragalactic objects increases because local measurement standards decrease**. In the documents, this relationship is expressed by a formula in which the proper or apparent distance depends on the comoving distance and an exponential factor associated with the Hubble constant and the time interval:

$$D(\Delta t) = D_o \cdot e^{(H_o \Delta t)}$$

In this formulation, D is the apparent distance measured by a terrestrial observer, D_0 is the comoving distance, and Δt is the time interval elapsed from the photon's emission to its reception.

The second proposition is that **redshift can be interpreted as the cumulative effect of the local contraction of measurement standards**. The article on distance as a function of redshift derives the relation:

$$1 + z = e^{(H_0 T)}$$

and, from it:

$$T = \ln(1 + z) / H_0$$

Since $T = D_0/c$, the comoving distance as a function of redshift is presented as:

$$D_0 = c \cdot \ln(1 + z) / H_0$$

and the proper or apparent distance as:

$$D = c(1 + z) \cdot \ln(1 + z) / H_0$$

These expressions constitute one of the mathematical bases of DUT for reinterpreting the relationship between redshift and cosmological distance.

The third proposition is that **galactic mass and age affect redshift**. In the appendices, the theory predicts that, holding distance and age constant, more massive galaxies should exhibit lower redshift, since greater gravitational intensity would produce greater contraction and, therefore, a shorter emitted wavelength. Similarly, holding mass and distance constant, older galaxies should exhibit lower redshift, since they would have accumulated more time of gravitational contraction.

The fourth proposition is that **effects attributed to dark matter could be reinterpreted as effects of differential gravitational contraction**. According to the documents, stars in regions farther from the galactic center are subjected to less intense gravitational fields, which would produce less contraction and a longer relative wavelength. This effect would be measured as greater redshift and interpreted, in the conventional model, as greater orbital velocity.

3. Methodology

The methodology adopted in the documents can be reconstructed on three levels: theoretical-deductive, mathematical, and observational.

3.1. Theoretical-deductive procedure

The first level consists of taking the single hypothesis of continuous gravitational contraction as a starting point. From it, the documents seek to deduce different phenomena: Hubble's Law, the distance-redshift relation, cosmological time dilation, galactic rotation curves, the Tully-Fisher relation, and correlations between mass, age, and redshift. This strategy seeks to show that a single premise could explain phenomena usually attributed to distinct mechanisms.

3.2. Mathematical procedure

The second level consists of formalization through exponential functions. Contraction is modeled by factors of the type:

$$F_i(\Delta t) = e^{-(H_0 \Delta t)}$$

or, in more general formulations, by factors dependent on mass (M), radial distance (R), and contraction time. In the article on the Tully-Fisher relation, for example, contraction is generalized with the additional hypothesis that the contraction time is inversely proportional to the intensity of the gravitational field. From this hypothesis, a factor dependent on M/R^2 is introduced.

The article on time dilation applies similar reasoning to units of time. If the unit of distance in a proper reference frame contracts as:

$$m_t = m_0 / e^{-(H_0 \Delta t)}$$

then the unit of time is also altered. From the physical definition of the second and the comparison between comoving and proper reference frames, the document derives:

$$S_t = S_0 / Z$$

and, for the numerical measure of time:

$$T_t = Z \cdot T_0$$

This relation is used to explain the cosmological time dilation observed in Type Ia supernovae.

3.3. Observational procedure

The third level is observational. The documents present three main types of tests.

The first test compares pairs of galaxies with similar approximate distance, different mass, and redshift greater than 0.01. The prediction is simple: at the same distance, the more massive galaxy should exhibit lower redshift. The document reports that 47 of 50 pairs, or 94%, obey this condition.

The second test uses galactic morphology as an approximation for age. Elliptical and lenticular galaxies are treated as older populations; spiral galaxies, as younger populations. In clusters such as Virgo, Coma, and Centaurus, the documents present average redshift differences in which elliptical/lenticular galaxies exhibit lower redshift than spiral galaxies.

The third test seeks to isolate variables within subgroups of the Coma Cluster. The sample is restricted to 20 elliptical/lenticular galaxies, with the goal of keeping distance and age approximately constant. The set is then divided into subgroups with a redshift variation of about 150 km/s, seeking to reduce the effect of peculiar velocities. DUT's prediction is a negative correlation between stellar mass and observed redshift.

4. Theoretical Model and Main Calculations

4.1. Distance, redshift, and photon travel time

The model starts from the relation:

$$D(\Delta t) = D_0 \cdot e^{(H_0 \Delta t)}$$

This equation expresses the idea that the apparent distance increases because the local measurement standard decreases. Applied to wavelength, the same logic leads to:

$$\lambda = \lambda_0 \cdot e^{(H_0 T)}$$

The definition of redshift is:

$$z = (\lambda - \lambda_0) / \lambda_0$$

Hence:

$$1 + z = e^{(H_0 T)}$$

and:

$$T = \ln(1 + z) / H_0$$

Since $D_0 = cT$, one obtains:

$$D_0 = c \cdot \ln(1 + z) / H_0$$

This equation is presented as the direct formula for comoving distance as a function of redshift. The apparent distance, in turn, is given by:

$$D = D_0(1 + z)$$

therefore:

$$D = c(1 + z) \cdot \ln(1 + z) / H_0$$

The central point is that, in DUT, redshift is treated not merely as an effect of spatial expansion, but as a consequence of the difference between contracted measurement standards and photons that traverse regions of negligible gravity over long time intervals.

4.2. Cosmological time dilation

The temporal application of DUT starts from the idea that the physical definition of the second depends on atomic standards, and that these standards would also be affected by gravitational contraction. The document on Type Ia supernovae distinguishes between comoving time (S_0) and proper time (S_t). If the contraction factor is:

$$Z = e^{(H_0 \Delta t)}$$

then the proper unit of time is:

$$S_t = S_0 / Z$$

Consequently, the numerical measure of a physical duration in the proper reference frame becomes:

$$T_t = Z \cdot T_0$$

This equation is used to interpret the cosmological time dilation of Type Ia supernovae. The article states that the observed duration of SN1A explosions grows by the factor associated with $(1 + z)$, which would be consistent with DUT's prediction.

4.3. Mass, age, and redshift

In the documents, redshift depends on at least three factors: distance, mass, and age. Distance increases redshift because it extends the photon's travel time; mass reduces redshift when the emitting galaxy is more massive, since greater gravity would imply greater contraction and a shorter emitted wavelength; age reduces redshift because older galaxies would have accumulated more contraction over time.

This proposition leads to two observational predictions:

$$M_1 > M_2 \Rightarrow z_1 < z_2$$

when the galaxies are at the same distance and have comparable age; and:

$$Age_1 > Age_2 \Rightarrow z_1 < z_2$$

when mass and distance are approximately controlled.

In the galaxy pairs presented in the appendices, 47 of 50 cases are classified as favorable to the prediction "greater mass, lower redshift."

In the morphology-based test, the averages presented are:

Cluster	Old type — E/S0	Young type — Spiral	Difference
Virgo	950 km/s	1450 km/s	+500 km/s
Coma	6900 km/s	7100 km/s	+200 km/s
Centaurus	3000 km/s	3500 km/s	+500 km/s

The proposed interpretation is that elliptical/lenticular galaxies, being older, would have accumulated greater contraction and would therefore exhibit lower average redshift. The document itself acknowledges, however, that the Λ CDM model offers an alternative explanation based on morphological segregation and peculiar velocities within clusters.

4.4. Galactic rotation curves

To explain rotation curves without dark matter, the documents propose that stars in different regions of a galaxy emit photons subjected to different contraction histories. The Newtonian orbital velocity is expressed as:

$$V = \sqrt{(MG/R)}$$

but the apparent velocity measured on Earth would also include a differential contraction factor. The general equation presented for the apparent rotation curve is:

$$V_t = c \cdot [((1/c) \cdot \sqrt{(MG/R)} + 1) \cdot e^{(H_0 T_0 - J_0 M T_0 / R^2)} - 1]$$

In this expression, c is the speed of light, M is the mass of the emitting galaxy, G is the gravitational constant, R is the distance from the star to the galactic center, H_0 is the Hubble constant, T_0 is the contraction time, and J_0 is the Joxaxian constant. The document applies this equation to the galaxy Messier 33 and states that the resulting curve reproduces the behavior traditionally attributed to dark matter.

4.5. Tully-Fisher relation

The article on the Tully-Fisher relation starts from the empirical relation:

$$M \propto V^4$$

According to the document, DUT allows this relation to be derived from the galactic rotation equation and from two astrophysical approximations for spiral galaxies: an approximately constant mass-to-light ratio and an approximately uniform central surface brightness, associated with Freeman's Law.

Assuming that the mass of a spiral galaxy is proportional to the area of its disk:

$$M = \rho \cdot \pi \cdot R^2$$

one obtains:

$$R = \sqrt{(M / (\pi \cdot \rho))}$$

Substituting this relation into the Newtonian form of velocity:

$$V = \sqrt{(MG/R)}$$

one obtains, in proportional terms:

$$V \propto M^{1/4}$$

hence:

$$M \propto V^4$$

The document concludes that the Tully-Fisher relation could be derived without recourse to dark matter, provided the approximations regarding surface density and mass-to-light ratio for spiral galaxies are accepted.

5. Results and Discussion

The results presented in the documents can be grouped into four sets.

The first set concerns the **distance-redshift relation**. DUT proposes direct formulas for comoving distance and apparent distance as functions of redshift. The strength of this formulation lies in its mathematical simplicity: the relation depends on c , H_0 , and $\ln(1 + z)$. The limitation lies in the fact that the document itself acknowledges that the formula would be more adequate for galaxies of mass similar to the Milky Way, and could be compromised when applied to much more or much less massive galaxies.

The second set concerns **cosmological time dilation**. The theory derives the relation $T_t = Z \cdot T_0$, applying it to the observed duration of Type Ia supernovae. The argument is consistent with DUT's internal logic, since time and space are treated as physical standards equally affected by gravitational contraction. However, the demonstration requires broader quantitative comparison with observational databases independent of SN1A light curves.

The third set concerns **galactic mass and age as redshift factors**. The tests presented are favorable to DUT in some cases: 47 of 50 galaxy pairs obey the condition "greater mass, lower redshift"; elliptical/lenticular galaxies exhibit lower average redshift than spirals in Virgo, Coma, and Centaurus; and subgroups of the Coma Cluster show negative correlations between stellar mass and redshift in two of the four groups analyzed.

The fourth set concerns **galactic dynamics**. DUT proposes that rotation curves and the Tully-Fisher relation can be derived without dark matter. The equation applied to Messier 33 and the derivation of $M \propto V^4$ constitute attempts to show that the theory reproduces important empirical regularities. The main limitation is that these derivations depend on auxiliary hypotheses — especially regarding surface density, mass-to-light ratio, and simultaneous galaxy formation — that need to be tested on larger samples.

The most methodologically careful test is that of the Coma subgroups. In it, the documents seek to control for distance, age, and radial velocity. The results, however, are mixed: subgroup 2 shows a correlation of -0.8477 ; subgroup 4 shows a correlation of -0.6789 ; but subgroup 1 shows a perfect positive correlation, albeit with only two galaxies, and subgroup 3 shows a weak positive correlation. The document itself acknowledges that the absence of a universal negative correlation indicates that the effect proposed by DUT is not the only factor involved.

The limitations made explicit are relevant: small subgroups, the use of stellar mass as an approximation of total mass, possible projection effects, and differences in depth within the cluster. For this reason, the most appropriate conclusion is not that DUT has been confirmed, but that its preliminary results justify further investigation and more robust mathematical formalization.

6. Conclusion

Based on the documents analyzed, the Decreasing Universe Theory can be formulated as a theoretical program based on a central hypothesis: gravity produces continuous contraction of space, matter, and local measurement standards. From this hypothesis, the texts construct alternative explanations for cosmological redshift, apparent distance, time dilation, galactic rotation curves, the Tully-Fisher relation, and correlations between mass, age, and redshift.

DUT's main contribution lies in its unifying ambition. Rather than separately invoking dark energy, dark matter, accelerated expansion, and local dynamical effects, the theory seeks to derive multiple phenomena from a single premise. Its fundamental equations — especially $D = D_0 e^{(H_0 \Delta t)}$, $1 + z = e^{(H_0 T)}$, $D_0 = c \cdot \ln(1 + z) / H_0$, and $T_t = Z \cdot T_0$ — mathematically organize this proposal.

The data presented in the appendices are compatible with some of the theory's predictions: more massive galaxies appear, in several pairs, with lower redshift; morphologically older galaxies exhibit lower average redshift in some clusters; and Coma subgroups show negative correlations between mass and redshift when certain variables are controlled. In addition, the documents propose an equation for rotation curves and a derivation of the Tully-Fisher relation.

However, DUT's scientific validity still depends on important advances. It is necessary to more fully formalize the relationship between gravitational field, mass, exposure time, and contraction. It is also essential to test the predictions on independent astronomical catalogs, with larger samples, homogeneous total-mass estimates, statistical control of peculiar velocities, and direct quantitative comparison with the Λ CDM model.

Thus, the most balanced conclusion is that DUT, as presented in the attached documents, constitutes an alternative cosmological hypothesis that is coherent in its internal formulation and endowed with testable predictions, but not yet sufficiently demonstrated to replace the standard cosmological model. Its immediate academic value lies in proposing a research agenda: transforming the hypothesis of continuous gravitational contraction into a fully specified mathematical model and submitting it to independent, replicable, and statistically robust observational tests.